INCH-POUND

MIL-P-29609
26 June 1995

MILITARY SPECIFICATION

POLYIMIDE RETAINER MATERIAL, POROUS NON-METALLIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers basic characteristics required for porous polyimide materials intended for use as instrument and thin section ball bearing retainers (cages), and methods of determining these characteristics.

1.2 Classification. The materials shall be furnished in the following types and forms as specified (see 6.2).

Form RM - Rods, molded

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116 - Methods of preservation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center, Code 41K200B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330
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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-831 - Test Reports, Preparation of.

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from DODSSP-Customer Service, Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Applications of copies should be addressed to the American Society for Quality Control, 310 West Wisconsin Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D2240 - Rubber Property-Durometer Hardness.
ASTM-D2873 - Porosity, Mercury Intrusion Porosimetry.
ASTM-D3951 - Packaging, Commercial.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

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3.1.1 Material. Material shall be polyimide, organic condensation polymers derived from the reaction between aromatic dianhydrides and aromatic diamines.

3.1.2 Construction. This material shall consist of powder mixed, cured and uncured, to obtain a specified pore size and pore volume. The material is formed through the isostatic molding of the cured and uncured milled polyimide powders.

3.1.3 Characteristics. The material shall have the following characteristic values. These values apply only to standard sizes of rods and tubes listed in Table I. The characteristic requirements for special sizes shall be specified (see 6.2).

3.1.3.1 Hardness. Material shall have a minimum hardness of 84 Shore D.

3.1.3.2 Apparent density. Material shall have an apparent density between 1.1 and 1.2 grams per cubic centimeter (g/cc).

3.1.3.3 Ring tensile strength. Material shall have a minimum ring tensile strength of 2000 pounds per square inch (psi).

3.1.3.4 Porosity. Material shall have pore diameter from 0.95 (140.4 pounds per square inch, absolute (psia)) to 1.25 microns (144.7 psia) at a corresponding intrusion volume from 0.08 to 0.10 cc/gram, the total intrusion volume to be 0.13 to 0.20 cc/gram inclusive at an intrusion pressure of 1500 psia. The shape of the intrusion curve shall be similar to that shown in Figure 1.

3.1.4 Material defects. Material shall be considered defective if any debris or more than five non-polyimide inclusions over 0.010 inch in any dimension are found in any diametrical cross section.

3.1.5 Color. The color varies and shall not be a cause for rejection.

3.1.6 Dimensions and tolerances.

3.1.6.1 Lengths. Standard and minimum lengths per diameter shall be as specified in Table I. At least 75 percent of the rods of any given shipment shall be standard length or longer and no more than 25 percent of a given shipment shall be supplied in lengths between the minimum and standard shipping lengths without prior approval.

3.1.6.2 Diameters. Diameters shall be specified (see 6.2) at the time of order. Standard sizes are covered in Table I.

3.1.7 Erosion resistance. The test specimen shall show no sign of degradation after a minumum of five complete cycles (see 4.5.8).

3.2 Qualification. Material furnished under this specification shall be authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.5 and 6.2).

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TABLE I. Standard rod diameter and length.

Standard (±0.0030) Diameter	Standard (±0.10) Length	Minimum (±0.10) Length 1/
0.6250	9.50	5.00
0.8750	8.75	4.50
1.5000	8.00	4.00
1.7500	7.00	3.75

1/ Minimum length to be shipped without prior customer approval.

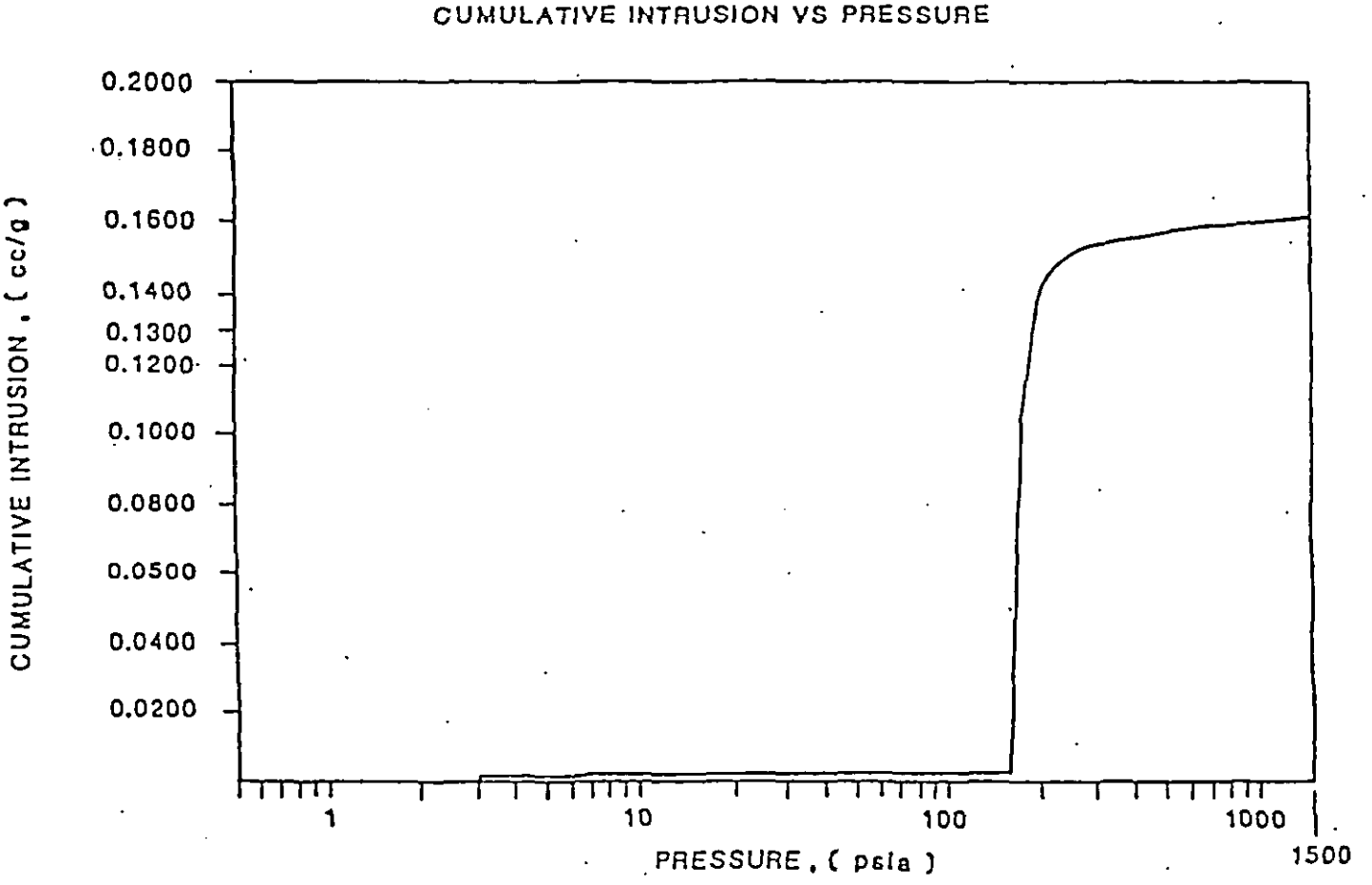


FIGURE 1. Intrusion curve.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be as specified in Table II.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in Table III.

4.5 Inspection requirements.

4.5.1 Inspection requirements. Examination of material shall be made in accordance with 4.5.2 through 4.5.9. Unless otherwise specified, the lot size shall be expressed in units of rods or tubes. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, form, size, cured fraction, and uncured fraction, from one distinct molding.

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TABLE II. Qualification inspection.

Inspection	Requirement Paragraph	Test Paragraph	Acceptable Quality Level
Hardness	3.1.3.1	4.5.2	0
Apparent density	3.1.3.2	4.5.3	0
Ring tensile strength	3.1.3.3	4.5.4, 4.5.4.1	0
Porosity	3.1.3.4	4.5.5, 4.5.5.1	0
Material defects	3.1.4	Visual	0
Lengths	3.1.6.1	4.5.6	0
Diameters	3.1.6.2	4.5.7	0
Erosion resistance	3.1.7	4.5.8	0

4.5.2 Hardness. The hardness shall be measured in accordance with ASTM-D2240 with the following exception. The hardness measurement may be made on the subject rod in lieu of preparing a test disc. The sample must meet the hardness requirements of 3.1.3.1.

4.5.3 Apparent density. The following method shall be used to confirm the values specified in 3.1.3.2.

a. Machine rod to finished dimensions.

b. Trim irregularities from end of rod.

c. Place rod in vacuum oven for 2 to 4 hours at 1-5 psia and 125±10°C.

d. Allow to cool to room temperature under a pressure of 1-5 psia.

e. When cool, measure weight (grams), diameter (inches), and length (inches). To determine the apparent density (see 3.1.3.2) use the following formula:

$$\text{Apparent Density (g/cc)} = \frac{\text{Weight}}{(\text{Diameter})^2 \times \text{Length}} \times 0.0777$$

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TABLE III. Quality conformance inspection.

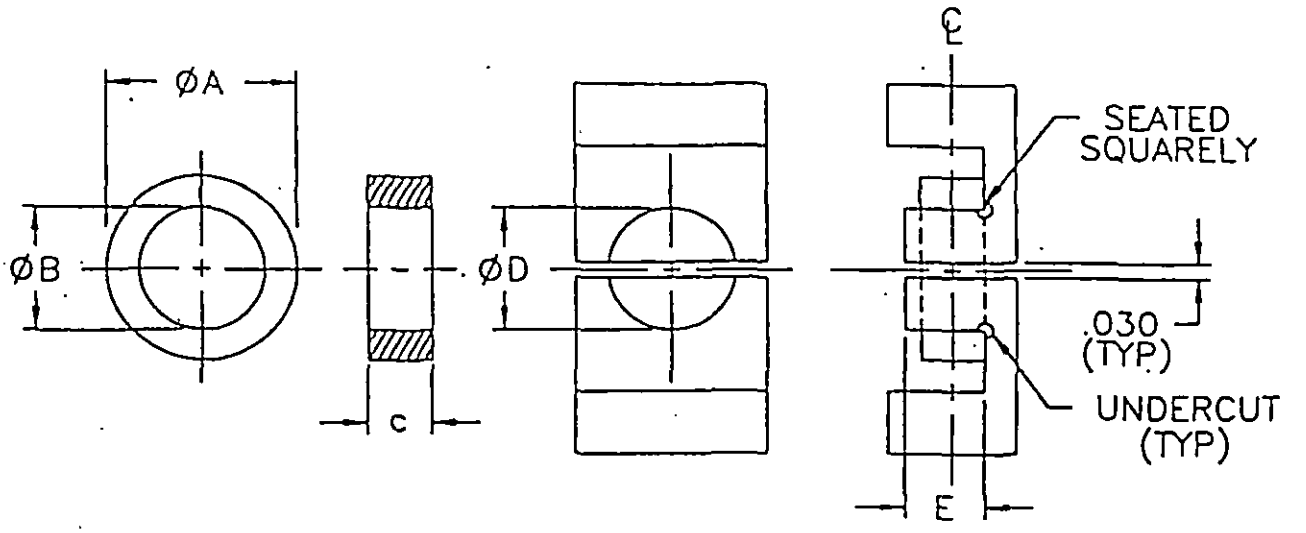
Inspection/ Characteristic	Requirement Paragraph	Test Paragraph	Quality Level
Hardness/minor			ASQC-Z1.4 1.5 AQL, Normal Inspec. Level II 1/ AC=0, RE=1
Apparent density/ minor	3.1.3.2	4.5.3	ASQC-Z1.4 1.5 AQL, Normal Inspec. Level II 1/ AC=0, RE=1
Rings tensile stength/major	3.1.3.3	4.5.4 4.5.4.1	2 rings, one rod per lot
Porosity/major	3.1.3.4	4.5.5 4.5.5.1	1 sample, one rod per lot
Material defects/ minor	3.1.4	Visual	100%
Lengths/minor	3.1.6.1	4.5.6	100%
Diameters/minor	3.1.6.2	4.5.7	ASQC-Z1.4 1.5 AQL, Normal Inspec. Level II 1/ AC=0, RE=1

1/ Lot size shall be restricted to 4-8 rods to ensure AC=0, RE=1.

Note: Do not handle rod with unprotected hands; oils from the hands are readily absorbed by the material and could cause inaccuracies in the density calculation.

4.5.4 Ring tensile strength. A minimum of two tensile specimens (see 3.1.3.3) will be machined from the screen end of one rod selected at random from the molded batch. Specimen dimensions will be according to the diameter of the rod (see Figure 2). All rings shall be tested at the uniform speed of 0.200 ± 0.050 inches/minute. The test fixture's split rod diameter shall be the nominal, ± 0.001 inches.

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TENSILE SPECIMEN

(LOADING TO BE ON ℄ .005)
TENSILE TOOLING

Bar Stock Size	ØA ±.005	ØB ±.001	C ±.005	ØD ±.00	E ± Min
Up to and including 3/4 dia.	.475	.312	.210	.312	.219
Over 3/4 dia. up to and including 1-1/4 dia.	.750	.500	.250	.500	.281
Over 1-1/4 dia.	1.250	.950	.360	.950	.375

FIGURE 2. Ring tensile strength.

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4.5.4.1 Calculation. Ring tensile strength (RTS) is the maximum stress sustained by the specimen during the test and is calculated as follows:

$$RTS = \frac{P}{C\ (\text{ØA}-\text{ØB})}$$

where P is the maximum load and ØA, ØB, and C are listed in Figure 2. ØA and ØB are diameters.

4.5.5 Porosity. Pore size distribution and pore volume (see 3.1.3.4) shall be measured using a mercury intrusion porosimeter in accordance with ASTM-D2873.

4.5.5.1 Procedure. A minimum of one sample per molding lot shall be tested. The sample shall be a 1/4 inch thick cross section of the rod taken at a point 1/2 inch from the plug end. The plug end is the end opposite the screen end. The rod is to be selected at random and tested as follows:

a. The sample is to be fractured into a single specimen approximately one gram in weight.

b. Weigh and record the sample weight to 0.0001 grams.

c. Conduct mercury intrusion porosimetry over a pressure range of 30 to 1500 psia, a mercury contact angle of 130°, and a mercury surface tension of 485 dynes/centimeter (cm).

d. Compare the data to the specification limits of 3.1.3.4. Typical trace and visual portrayal of 3.1.3.4 is depicted in Figure 1.

Failure of this sample to meet 3.1.3.4 will necessitate reinspecting the molding lot per ASQC-Z1.4, 1.5 AQL, normal inspection, Level II, AC=0, RE=1, before acceptance.

4.5.6 Length. As specified in 3.1.6.1, length shall be measured to 0.10 inch. The length shall be the shortest axial dimension of the rod taking into account angled or chipped ends.

4.5.7 Diameter. As specified in 3.1.6.2, diameter shall be the maximum diameter measured to the nearest 0.001 inch.

4.5.8 Erosion resistance. As specified in 3.1.7, erosion resistance shall be determined on one slug, cut from the plug end of every rod in the lot. The slug shall be 1/2 to 3/4 inch thick. It shall be polished on polishing wheels until smooth, scratch-free, and flat. Polished slugs shall be tested for erosion resistance as follows:

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a. The slugs shall be water impregnated at 24 millimeters (mm) of Hg for 10±2 minutes.

b. The slugs shall then be ultrasonically cleaned in water at 200 Cavons for 10±2 minutes using Branson E-Module or equivalent at full power of 40 Kilohertz (KHz).

c. The water shall be discarded and the slugs shall be vacuum dried at 80±5°C for 30±5 minutes.

d. Repeat b and c.

e. The slugs shall be visually inspected at 30X for erosion. If no erosion is detected, return to step (a). This represents one cycle of the five cycles minimum. If erosion is present, discontinue testing. Molding lot is unacceptable.

4.5.9 Inspection of preparation for delivery. Sample packages and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance to the requirements of Section 5.

4.6 Retention of qualification. To retain qualification the manufacturer shall forward a report to the qualifying activity in accordance with MIL-STD-831 at 24 month intervals. The report shall consist of the information specified in 4.3 (see 6.3 and 6.4).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Material to be packaged in accordance with ASTM-D3951.

5.2 Marking. In addition to any specific marking required by the contract or order, the interiors and exteriors of the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Polyimide produced to this specification is intended for instrument and thin section ball bearing retainers (cages).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

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c. Form material data required with each order.

(1) Nominal diameter.

(2) Total inches per lot.

(3) Lot number.

(4) Rod identification.

(5) Lot porosimetry data.

(6) Lot average ring tensile strength.

(7) Lot average hardness, Shore D.

(8) Lot average apparent density.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List 29609 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Commander, Naval Air Systems Command (AIR-4.1.9), 1421 Jefferson Davis Highway, Arlington, VA 22243-4190, and information pertaining to qualification of products may be obtained from that activity.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 227.405-70 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.6	DI-NDTI-80809A	Test/Insp Report	Paragraph 10.2.7

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements

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Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.5 Suggested source.

Furon, Advanced Polymers Division
Dixon Facility
386 Metacom Avenue
Bristol, Rhode Island 02809

6.6 Subject term (key word) listing.

Bearings
Cage
Separators

Custodians:
Army - MI
Navy - AS
Air Force - 84

Review activities:
Navy - CH

Preparing activity:
Navy - AS

(Project 9330-1174)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.
   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-29609	2. DOCUMENT DATE (YYMMDD) 950626

3. DOCUMENT TITLE
Polyimide Retainer Material, Porous Non-Metallic

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-2970 (2) DSN: 624-2970
c. ADDRESS (*Include Zip Code*) CODE 4.1.11.2B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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